

Demonstration of Directional Force Feedback in Virtual Reality Tennis using Dual-Flywheel Haptics

Allyson E. Chen^{1*}, Xuan Gedney^{1*}, and Jasmine Roberts^{1*}
UC San Diego

This demo introduces a haptic feedback device for tennis simulation, designed to provide realistic physical sensations of ball impact through interaction with a virtual racket model. The system integrates with a Unity-based virtual setup where users are visually immersed in a virtual tennis court environment. Unlike previous systems, this device incorporates a mechanical interface that replicates the sensation of a tennis ball striking the racket using a dual flywheel system.

This approach follows a one-controller, one-function paradigm, where each input device is dedicated to a specific function rather than serving as a general-purpose controller. This targeted design enhances realism and improves user immersion by providing precise, specialized feedback rather than distributing multiple functionalities across a single device. Furthermore, this system aligns with augmented virtuality, as it brings physical sensations into an otherwise fully virtual environment, reinforcing the sense of realism and presence.

1 INTRODUCTION

Haptic devices enhance virtual experiences by providing tactile feedback that mimics real-world sensations. In tennis, haptic technology offers opportunities to bridge the gap between physical and virtual environments. Earlier gaming innovations, such as the Wii tennis racket, introduced basic vibration feedback but lacked nuanced sensations of impact that vary with swing speeds and ball interactions [1]. Similarly, the PlayStation VR introduced motion and aim controllers, which laid the foundation for more immersive interactions, while PlayStation's DualSense controllers incorporated advanced haptic feedback for richer tactile experiences. However, these general-purpose controllers remain limited in delivering sport-specific feedback [2].

Our system addresses these limitations by creating a virtual tennis court environment that provides realistic haptic feedback. The device utilizes a dual flywheel system to simulate the sensation of ball impact, offering precise and varied feedback based on swing speed and ball interaction [3]. Figure 1 compares previous racket attachments, such as the Nintendo Wii and Meta Quest 3 designs, which relied on nested controllers, with our specialized haptic racket system. This approach bridges the gap between physical and virtual tennis experiences, providing a more immersive and authentic simulation.

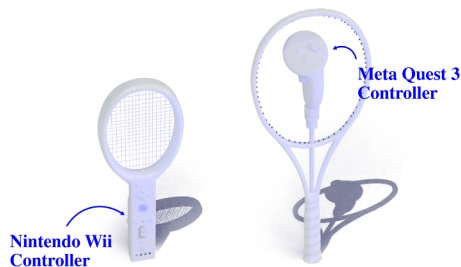


Figure 1: Nintendo Wii racket attachment (left) and the Meta Quest 3 racket attachment (right)

2 DESIGN AND IMPLEMENTATION

2.1 Haptic Racket Design

As shown in Figure 2, the haptic racket was designed to closely replicate a real racket's feel while accommodating internal components. Key specifications include **Total mass: 333g (standard: 280–300g)**, **Center of mass: 360mm from the base (standard: 340mm)**, and **Grip size: 1 (104.7 mm)**. The internal components include **Arduino Nano ESP32, motor driver, and MPU-6050 IMU**. The racket also features a **Dual flywheel system: 10:1 micro-metal gearmotors**.

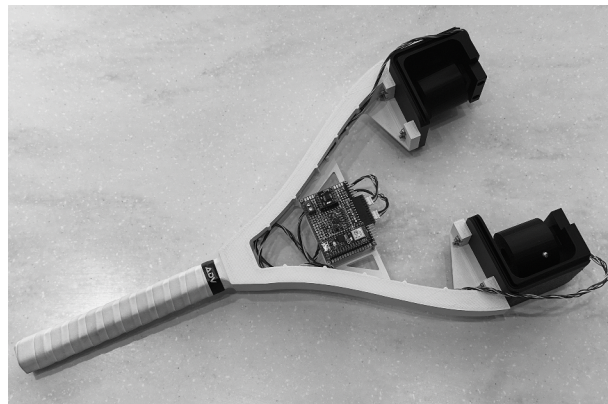


Figure 2: Haptic racket prototype with dual flywheels and embedded microcontroller for tennis simulation.

The internal components include a proto-board housing the microcontroller and sensors, and two flywheel-gimbal systems located on custom platforms at the racket arms. Each flywheel has an outer radius of 48mm, inner radius of 18mm, and height of 40mm.

2.2 Virtual Environment

A virtual tennis training environment was rendered in Unity, designed to engage users' visual and audio sensory experience. The environment allows users to enter a custom tennis stadium and interact with a virtual racket mapped to the haptic racket. Audio cues are relayed to the user through the virtual reality headset when the ball is struck.

2.3 System Architecture

The system integrates Unity I/O with an Arduino Nano to control the dual flywheel mechanism, illustrated in Figure 3. Unity manages the virtual ball simulation and collision detection, while the Arduino Nano processes inputs to provide real-time haptic feedback through the dual flywheel system.

¹ {aechen, xgedney, jar053}@ucsd.edu

* All authors contributed equally to this work

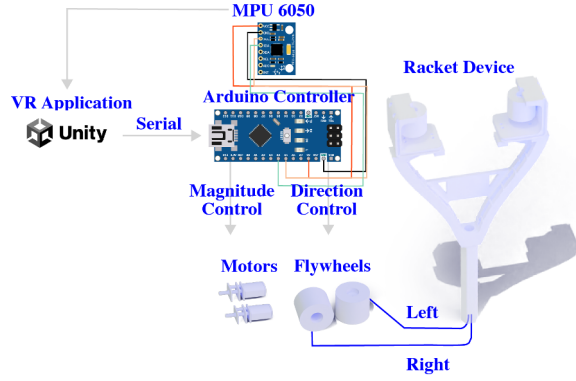


Figure 3: System architecture showing Arduino controller interfacing with MPU 6050 IMU for directional and magnitude control, connected to motors and flywheels for generating directional forces.

Unity releases a tennis ball every 10 seconds at regular intervals and detects racket-ball collisions. When a collision occurs, Unity calculates a simplified event signal and sends it to the Arduino Nano via serial communication. The Arduino then activates the dual flywheel system to provide physical sensation of ball impact. The mathematical model for the moment generated by the flywheels is given by:

$$M = -2I_{zz}\omega \quad (1)$$

Where I_{zz} is the moment of inertia and ω is the angular velocity. The output moment intensity is determined using IMU acceleration data, with motor spin time varying based on acceleration magnitude (Table 1).

Table 1: Acceleration magnitudes to motor spin time

Acceleration	Spin Time (ms)
20000	200
15000	175
10000	125
6000	50
5000	25
4000	10

3 RESULTS AND DISCUSSION

This paper presents a novel haptic feedback system for tennis simulation, designed to deliver realistic tactile sensations through a specialized dual flywheel mechanism (Figure 4). Preliminary evaluations highlight the system's ability to provide clear directional feedback and enhance user immersion, paving the way for future studies. User evaluations will further validate its effectiveness and potential applications in both gaming and training contexts.

During system development, several challenges occurred. The MPU 6050 produced noisy gyroscopic and accelerometer data, requiring additional filtering techniques. There were misalignments between the IMU and Unity disrupted depth perception, particularly along the Z-axis. Attaching a Quest controller resolved this issue, ensuring accurate positional data. Controlled tests have demonstrated that users could reliably identify active flywheels without visual cues, demonstrating its potential to provide clear directional feedback [3].

Future iterations will refine the dual flywheel system by integrating servos for continuous rotation and employing larger motors with higher torque. These upgrades will enable heavier flywheels for stronger and more realistic force production.



Figure 4: Close-up of the haptic racket integrated with the virtual environment.

4 CONCLUSION

This paper presents a novel haptic feedback system for tennis simulation, designed to deliver realistic tactile sensations through a specialized dual flywheel mechanism. Preliminary evaluations highlight the system's ability to provide clear directional feedback and enhance user immersion, laying the groundwork for future studies. Formal IRB-approved user evaluations will further validate its effectiveness and potential applications in both gaming and training contexts.

The device serves as a proof of concept for leveraging custom controllers in gaming and training paradigms. By focusing on a *one-controller, one-function* design, the system addresses the limitations of general-purpose VR controllers. Additionally, its potential integration into augmented reality environments positions it as a bridge between physical and virtual worlds.

Future work will prioritize refining evaluation metrics, and optimizing the hardware for improved force feedback. The exploration of mixed reality applications further underscores the potential of custom devices in reshaping immersive experiences across gaming, sports, and education.

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